Charles Pereira

Information Security Analyst

✉ chores265@hotmail.com

🔗 linkedin.com/in/charles-cyber-security/

Document: Security Assessment Report

Focus: Mobile Device Management (MDM)

Executive Summary: Technical assessment and audit of mobile devices in a corporate environment. The project applied compensating controls to ensure confidentiality and legal compliance (LGPD/GDPR) for data in transit and at rest for remote workers.

MDM | LGPD / GDPR | BYOD | Cryptography (HTTPS) | Incident Response

1. SITUATION (CONTEXT)

The organization faced a high risk of sensitive data exposure and potential sanctions from data protection authorities. The fleet of smartphones operated by employees lacked standardized protection policies. The devices transmitted data to organizational systems (Web Services) using unsecured protocols (HTTP) and locally retained strategic information in cache, creating a vast attack surface in scenarios of theft and network interception.

2. TASK (OBJECTIVE)

Act on the mapping and immediate remediation of vulnerabilities on employee devices. The objective was to audit connections, protect local storage, and establish a remote control framework based on Mobile Device Management (MDM) to ensure the regulatory compliance of devices before granting corporate access.

3. ACTION (TECHNICAL EXECUTION)

The execution involved the centralized orchestration of security policies through the MDM platform, applying the following restrictive measures:

- **Threat Identification and Triage:** Batch scanning to identify devices without antivirus protection and with communication protocols vulnerable to interception (Man-in-the-Middle).
- **Encryption in Transit Enforcement:** Blocking HTTP requests and mandating the HTTPS protocol, adding a security layer with SSL/TLS certificates for requests to Web Services.

- **Residual Data Management and Privacy:** Configuration of periodic clearing policies for Cache and Cookies, preventing confidential data from inappropriately residing in the device's browser.
- **Endpoint Control and Incident Response:** Forced installation (via push) of antivirus solutions (Endpoint Protection) and enabling location tracking with remote lock and wipe capabilities in case of loss or theft.



Fig 1. Triage highlighting encryption and antivirus failures.

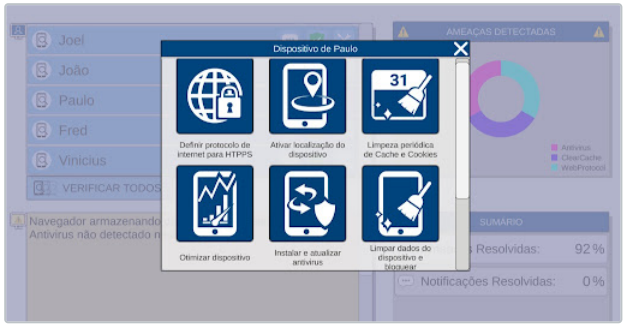


Fig 2. Remote mitigation: HTTPS, Antivirus, and Wiping via MDM.

4. RESULTS ACHIEVED

The active vulnerability rate in the fleet was reduced to 0%. Through standardization via MDM, the organization achieved full adherence to LGPD/GDPR principles, ensuring the pillars of Information Security (Confidentiality, Integrity, and Availability). Remote containment ensured that even in physical device theft scenarios, organizational data could not be extracted or reused by criminals.